YASH VARDHAN KUMAR

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EDUCATION

Maharaja Agrasen Institute of Technology, New Delhi *2023 – 2027*
B.Tech in Electronics and Communication Engineering

EXPERIENCE

Robotics Intern — Botivity Technologies LLP *Jun 2025 – Aug 2025*

- Configured PID loops for stable GPS waypoint tracking on an autonomous UGV (Pixhawk + Jetson Orin Nano).
- Built ROS 2 nodes for mission control and sensor integration; reduced command latency via MAVROS over UART.
- Fused LiDAR, Intel RealSense, and GPS for terrain mapping and obstacle avoidance; implemented GPS-loss and power-anomaly failsafes.
- Conducted field validation under varied GPS and environmental conditions with routine calibration and battery management.

PROJECTS

HOMIE Teleop Glove — Wearable Hand Teleoperation System *Jan 2026 – March 2026*

- Replicate the HOMIE GLOVES using their cad
- Wrote firmware on ESP32 (ESP-IDF); integrated micro-ROS to publish live joint states as a ROS 2 node, driving the gripper to mirror the operator's hand movements.
- System produces paired operator–robot joint trajectories suitable as demonstration data for imitation learning and embodied AI training pipelines.

Tech Stack: *ESP32, ESP-IDF, micro-ROS, ROS 2, HOMIE (open-source), Unitree G1*

Krishi Cobot — Agricultural Cobot System (e-Yantra IIT Bombay) *Sep 2025 – Feb 2026*

- Built a farm automation system: differential-drive robot navigates the field while a stationary UR5 arm handles manipulation at a fixed station.
- Developed CV pipeline (ROS 2 + OpenCV) to detect and classify fruit quality in a mixed tray using shape and colour features.
- Programmed UR5 to read ArUco markers on both the fertiliser can and the differential-drive bot to determine pick-and-place targets; deposits bad fruit into a bin.
- Implemented LiDAR-based 2D shape detection on the ground to trigger contextual behaviour changes during navigation.

Tech Stack: *ROS 2, Python, OpenCV, ArUco, LiDAR, Gazebo, UR5*

AndhaDhun — Haptic Gaming Controller for the Visually Impaired *May 2025*

- Built handheld ESP32 controller (ESP-IDF) with 8-zone spatial audio (OpenAL) and haptic feedback; custom I²C IMU driver for real-time orientation tracking.
- Integrated micro-ROS for gameplay sync; haptic response triggers when IMU aim aligns with a target zone.

Tech Stack: *ESP32, ESP-IDF, micro-ROS, I²C, IMU, OpenAL*

SSH over USB — Raspberry Pi Zero v1 (Buildroot) *April 2026 – Present*

- Enabled SSH-over-USB on Pi Zero v1 (no onboard network) with a custom Buildroot image — configured USB gadget mode (g_ether) and dropbear SSH in the kernel/rootfs build.

Tech Stack: *Raspberry Pi Zero, Buildroot, Linux, USB Gadget (g_ether)*

IK Controller — FreeRTOS Omnidrive Bot *March 2026 - Present*

- Implemented an inverse kinematics controller in FreeRTOS on ESP32 for a 4-wheel omnidirectional drive

Tech Stack: *ESP32, FreeRTOS, PlatformIO, C++*

SKILLS

Languages: Python, C, C++
Robotics & Embedded: ROS 2, micro-ROS, MAVROS, ArduPilot, ESP-IDF, FreeRTOS, PlatformIO
Simulation & Tools: Gazebo, RViz, ArduPilot SITL, Fusion 360, Buildroot, Git, Linux

ACHIEVEMENTS

- 1st place — Autonomous Drone Competition, Cognizance 2025, IIT Roorkee
- Semi-finalist — e-Yantra Robotics Competition 2025–26, IIT Bombay